Network Manager Release Notes

Version 2.2

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Contents

1: Introduction 4
1.1 Intended Audience 4
2: Installation 5
2.1 Prerequisites 5
2.2 Installing the Software 5
2.3 Creating a New Database 5
2.4 Upgrading an Existing Database 6
3: Functional Changes 7
3.1 Tick Mark Support Added 7
3.2 Linear Assembly Support Added 7
3.3 Labor Costs 7
3.4 Fiber Loss 8
3.5 Reporting 9
3.6 Trace Schematic 10
4: End-user GUI Changes 11
4.1 Cable Tree 11
4.2 Spec Manager 11
5: Administrator GUI Changes 13
5.1 Comms Settings Page 13
5.2 Advanced Settings Page 13
6: Data Model Changes 14
6.1 Labor costs 14
6.2 Fiber Loss settings 14
6.3 New design rights 14
6.4 Tick marks 14
6.5 Spec retired fields: 15
6.6 Fiber cable spec color scheme 15
6.7 Circuit model refactored 15

7: API Changes 16
7.1 JavaScript API 16
7.2 Circuit Manager 16
7.3 REST API 16
8: Documentation Changes 17
9: Upgrading 18
9.1 Upgrading Your Database 18
9.2 Post-upgrade Actions 18

1: Introduction

This document provides a summary of the major changes in this release of Network Manager for Telecoms.

1.1 Intended Audience

This document is intended for Administrators and Developers with a detailed technical knowledge of IQGeo Platform and utilities.

2: Installation

The following changes relate to installation and upgrade.

2.1 Prerequisites

Component	Version	Comments
IQGeo Platform	6.2	Mandatory
IQGeo Workflow Module	6.2	Mandatory
IQGeo Anywhere	6.2.0 (or higher)	Optional

All components must include any released patches.

2.2 Installing the Software

1. Uncompress the software download and copy the **comms** folder into:

   ```
   <myWorld_install_dir>/WebApps/myworldapp/modules
   ```

2. Download the module's JavaScript dependencies:

   ```
   myw_product fetch node_modules
   ```

3. Rebuild your JavaScript bundles:

   ```
   myw_product build applications
   ```

2.3 Creating a New Database

If you do not already have an IQGeo database, or that database does not yet have the Network Manager tables installed, you will need to:

- Install the Network Manager tables.
- Configure the network object types for your installation.

These processes are covered in detail in the *Installation and Configuration Guide*.

2.4 Upgrading an Existing Database

If you already have a Network Manager database from a previous release, you will need to:

- Upgrade your database.
- Perhaps perform some post-upgrade configuration changes.

These processes are covered in detail in *Upgrading* on page 18.

3: Functional Changes

3.1 Tick Mark Support Added

Cable distance markings, aka 'tick marks' can now be created, updated and displayed in the system. If set, cable's length will be derived from tick marks.

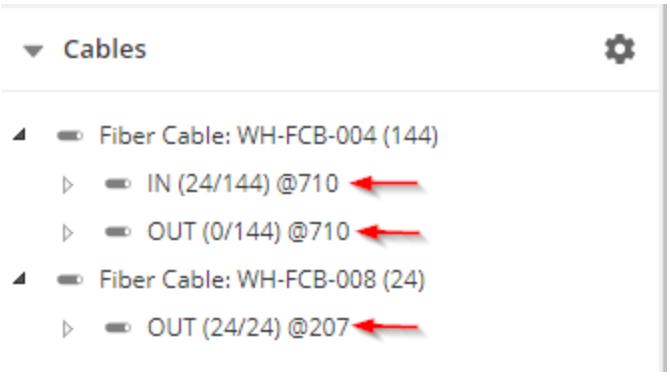


Figure 3-1: Tick mark display in cable tree

3.2 Linear Assembly Support Added

Network Manager now supports the ability to create and use linear assemblies from the structure palette. Examples of linear assemblies include:

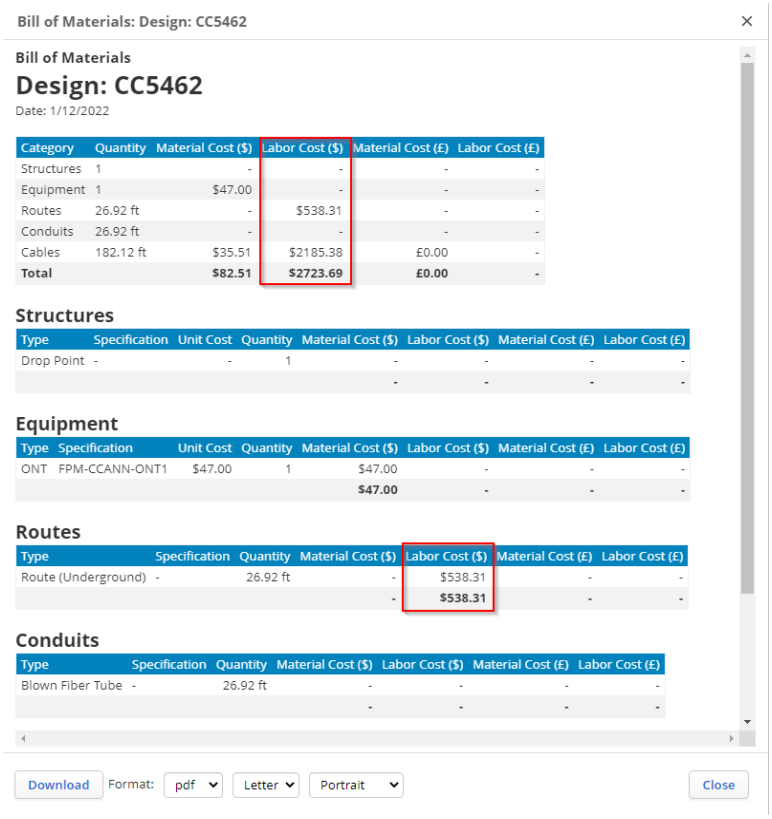
- Underground Route with conduit
- Underground Route with conduit housing cable
- Underground Route with nest blown fiber tubve
- Overhead Route with drop cable

3.3 Labor Costs

The Bill of Material (BOM) report has been extended to calculate cost of labor. Most objects can have labor cost associated during the design process.

- Object specification records can now include a global labor cost applied to new objects with the spec.

- Individual objects can override the spec derived cost .
- The product now includes a Labor Cost Manager, a catalog of labor costs



Bill of Materials: Design: CC5462

Bill of Materials

Design: CC5462

Date: 1/12/2022

Category	Quantity	Material Cost ($)	Labor Cost ($)	Material Cost (£)	Labor Cost (£)
Structures	1	-	-	-	-
Equipment	1	$47.00	-	-	-
Routes	26.92 ft	-	$538.31	-	-
Conduits	26.92 ft	-	-	-	-
Cables	182.12 ft	$35.51	$2185.38	£0.00	-
Total		**$82.51**	**$2723.69**	**£0.00**	**-**

Structures

Type	Specification	Unit Cost	Quantity	Material Cost ($)	Labor Cost ($)	Material Cost (£)	Labor Cost (£)
Drop Point	-	-	1	-	-	-	-
				-	**-**	**-**	**-**

Equipment

Type	Specification	Unit Cost	Quantity	Material Cost ($)	Labor Cost ($)	Material Cost (£)	Labor Cost (£)
ONT	FPM-CCANN-ONT1	$47.00	1	$47.00	-	-	-
				$47.00	**-**	**-**	**-**

Routes

Type	Specification	Quantity	Material Cost ($)	Labor Cost ($)	Material Cost (£)	Labor Cost (£)
Route (Underground)	-	26.92 ft	-	$538.31	-	-
			-	**$538.31**	**-**	**-**

Conduits

Type	Specification	Quantity	Material Cost ($)	Labor Cost ($)	Material Cost (£)	Labor Cost (£)
Blown Fiber Tube	-	26.92 ft	-	-	-	-
			-	**-**	**-**	**-**

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Figure 3-2: BOM reprot with labor cost

3.4 Fiber Loss

Fiber loss along a cable and at equipment is now calculated as part of a fiber trace. It is displayed in the trace results hover, the fiber trace schematic and the fiber trace report.

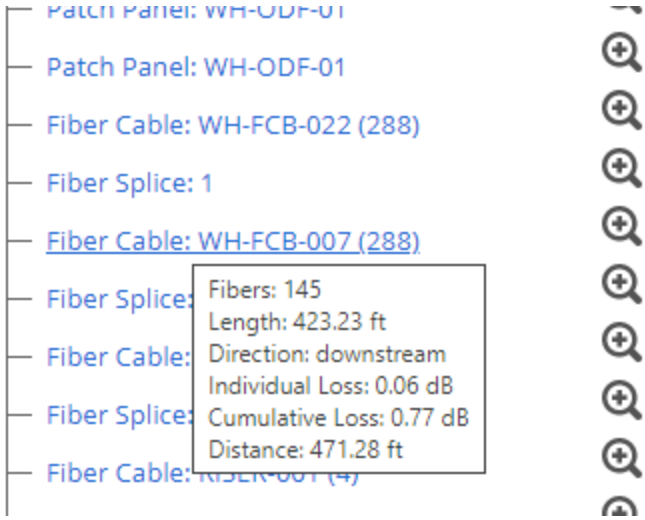


Figure 3-3: 4-1 Fiber Loss values in Trace Result hover

3.5 Reporting

Circuit Report

Fiber Cable: WH-FCB-022 (288)

Date: 1/12/2022

Cable	Fiber	Name	Service Type	Status	Customer
WH-FCB-022	#001	WH-FTTH-030	PON1	In Service	Jennings,M
	#001	WH-FTTH-031	PON1	In Service	Middleton,A
	#001	WH-FTTH-032	PON1	In Service	Morton,LL
	#001	WH-FTTH-033	PON1	In Service	Henshawe,B
	#001	WH-FTTH-034	PON1	In Service	Ellison,M
	#001	WH-FTTH-035	PON1	In Service	Davies,D
	#001	WH-FTTH-036	PON1	In Service	Careys,T
	#002	WH-FTTH-037	PON1	In Service	Brandon,E
	#002	WH-FTTH-038	PON1	In Service	Goulding,W
	#002	WH-FTTH-039	PON1	In Service	Harrington,H
	#002	WH-FTTH-040	PON1	In Service	Lucas,SW
	#002	WH-FTTH-041	PON1	In Service	Morris,M
	#003	WH-FTTH-042	PON1	In Service	Nicholls,M
	#003	WH-FTTH-043	PON1	In Service	Pratt,M
	#003	WH-FTTH-044	PON1	In Service	Bertram,J
	#003	WH-FTTH-045	PON1	In Service	Bertram,ST
	#003	WH-FTTH-046	PON1	In Service	Norris,M
	#004	WH-FTTH-047	PON1	In Service	Price,S
	#004	WH-FTTH-048	PON1	In Service	Bragge,M
	#004	WH-FTTH-049	PON1	In Service	Mitchell,F
	#005	WH-FTTH-051	PON1	In Service	Manwaring,M
	#005	WH-FTTH-052	PON1	In Service	Otway,G
	#005	WH-FTTH-054	PON1	In Service	Thorpe,J
	#005	WH-FTTH-055	PON1	In Service	Coucy,RD
	#005	WH-FTTH-056	PON1	In Service	Martin,SJ
	#006	WH-FTTH-057	PON1	In Service	Vernon,C
	#006	WH-FTTH-058	PON1	In Service	Vernon,F
	#006	WH-FTTH-059	PON1	In Service	Scroggs,M
	#007	WH-FTTH-050	PON1	In Service	Nash,M
	#007	WH-FTTH-053	PON1	In Service	Morland,M

Figure 3-4: Reporting

The system has added the following reports:

Fiber Report: Shows all fibers (and circuits) on a cable or on all cables within a route

Circuit Report: Shows all circuits on cable (or all cables within a route)

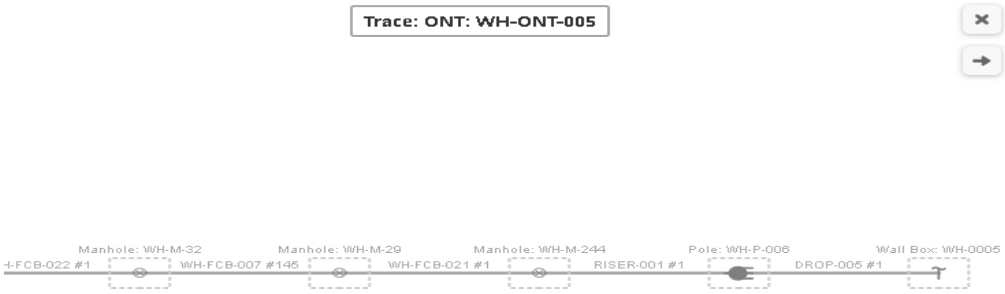
Equipment Circuit Report: Shows all circuits within an equipment (or all equip in a structure)

Spec Report: Shows all specs in the system or the selected spec

3.6 Trace Schematic

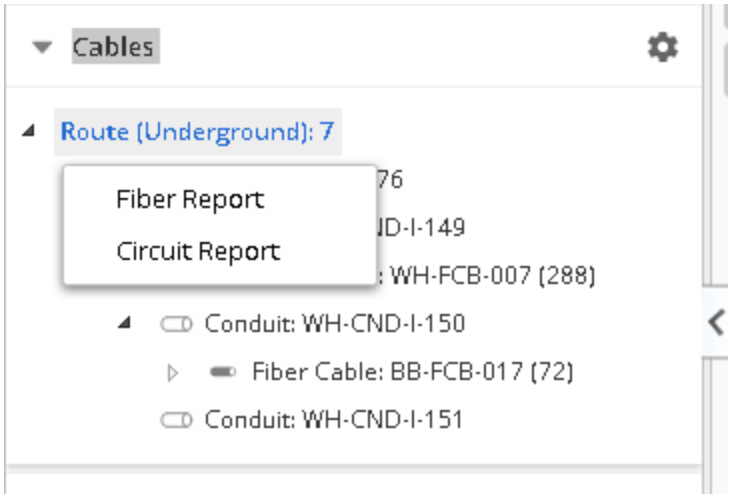
The trace schematic has been improved at this release.

- It now displays the structures equipment is housed in.
- It now has the option to show/hide labels by cables, equipment and/or structures
- If the schematic is open, it is now included in the fiber trace/trace reports



4: End-user GUI Changes

4.1 Cable Tree



The **Cable Tree** has been enhanced as follows:

- The context menu for route and cable nodes includes new entries Fiber Report and Circuit Report.

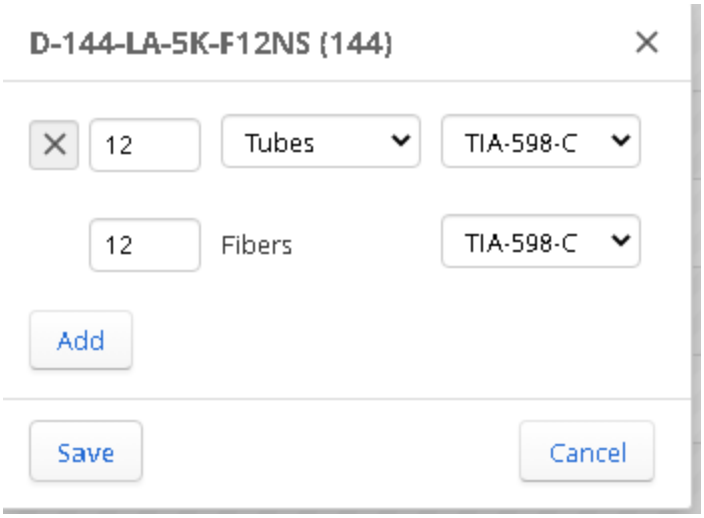
For more information on the cable tree, see the *User Guide*.

4.2 Spec Manager

The Spec Manager has been enhanced as follows:

- There is a new button Show spec report that generates a report for the spec category.
- For fiber cable specs the cable structure field has been improved describe the structure of the cable.

- For fiber cable specs the cable structure editor has been improved to be able to define the color scheme used at each level of the cable structure



For more details on the spec manager, see the *User Guide*.

5: Administrator GUI Changes

5.1 Comms Settings Page

The Comms tab in the Configuration -> Settings page has been modified as follows:

1. Labor Cost: There is a new tab for registering features with labor costs.
2. Fiber Color Schemes: Custom fiber color scheme engines are no longer required and are therefor no longer part of the configuration. The UI for choosing colors of a defined color scheme has been improved. It is now possible to define striped colors in the color scheme color chooser.

5.2 Advanced Settings Page

Thereis a new settings for configuring fiber loss constants.

	Purpose
mywcom.fiberLoss	Defines the amount of fiber signal loss at equipment of certain functions and at distance along a cable in decibels (dB)

For more information, see the *Installation and Configuration Guide*.

6: Data Model Changes

6.1 Labor costs

A new system-level feature has been added : mywcom_labor_cost, it is a member of the mywcom_ structures layer.

A new advanced setting has been added: mywcom.laborCosts

A new right has been added: mywcom.manageLaborCosts. All roles with the right mywcom.manageSpecifications have been granted this right as part of the upgrade.

All registered physical Network Manager Comms objects (structures, routes, equipment, cables & conduits) and specs have a new field added: *labor_cost*. If desired, this field can be removed post upgrade, so long as the change is reflected in the Labor Cost config section.

6.2 Fiber Loss settings

A new advanced setting is available mywcom.fiberLoss for configuring global fiber loss values (in db) for certain equipment functions and along cable distance.

The advanced platform setting core.units has been given a new member:

```
"fiber_loss": {
  "base_unit": "dB",
  "units": {
    "dB": 1
  }
```

6.3 New design rights

A new right has been added to control what roles can publish designs: mywcom.publishDesign All roles with the core.editFeature right have been granted this right as part of the upgrade.

6.4 Tick marks

The system level mywcom_fiber_segment feature has two new fields: *in_tick*, *out_tick*

All registered cable features and cable specification features have a new field: *tick_mark_spacing*

6.5 Spec *retired* fields:

Specifcation features' field *retired* is now mandatory to remove any nullish state.

6.6 Fiber cable spec color scheme

All registered fiber cable spec features have new field: *cable_structure* which replaces the fields *color_scheme* and *color_scheme_config*The contents of the old fields have been migrated to the new field as part of the database upgrade.

6.7 Circuit model refactored

The way circuits are modeled in the system has been completely refactored. This refactor drastically reduces the size of the database and offline extracts and will therefor scale much better.

The database upgrade removes two system level features and any references to them: mywcom_circuit_segment and mywocm_circuit_port.

7: API Changes

7.1 JavaScript API

The supported JavaScript API has been modified as follows:

1. FiberColorEngine:Custom fiber color engine classes are no longer required,. They can be removed. Fiber color schemes can now be created via configuraiton and applied via the spec manager's cable structure editor. Custom fiber color scheme definitions have been migrated as part of the data base upgrade.

For more details on these changes contact IQGeo support.

7.2 Circuit Manager

The server side circuit_manager class has been rewritten. If custom code is accessing methods from there they will need to be modified.

For more details on these changes contact IQGeo support.

7.3 REST API

There are no changes to the supported REST API at this release.

8: Documentation Changes

The documentation set has been updated for the changes described above.

All documentation can be found in the *public/doc* directory.

The documentation is published in a responsive HTML format that adapts to the user's device and facilitates searching across guides.

9: Upgrading

This section provides an overview of the upgrade process for this release.

9.1 Upgrading Your Database

To upgrade from the previous release of Network Manager:

1. Apply all IQGeo Platform upgrades:

```
myw_db <db_name> upgrade core 620
```

2. Apply the Network Manager upgrade:

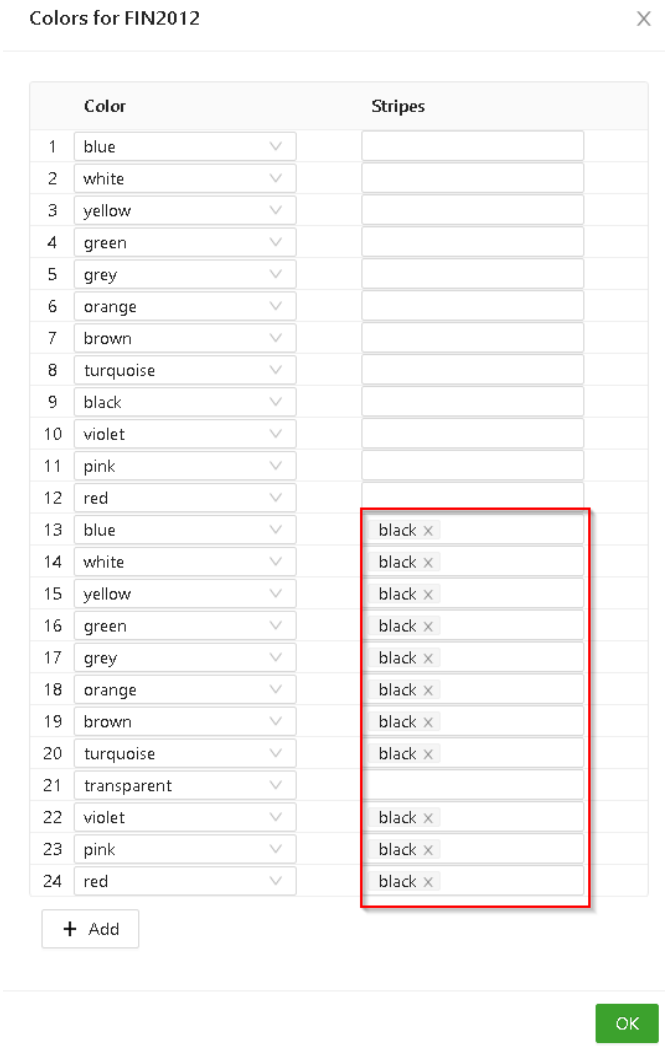
```
myw_db <db_name> upgrade comms 6222
```

3. Perform the post upgrade actions listed below.

9.2 Post-upgrade Actions

After upgrading your database, you should check the following:

1. Custom fiber color schemes that make use of striping must manually add the stripes in the Comms Settings at Comms -> Fiber Color Schemes -> Colors



You should also review any custom code for impact from the changes described in *API Changes* on page 16.

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